



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-26
HALF YEARLY EXAMINATION

CLASS: X
SUBJECT: CHEMISTRY [SET A]

FULL MARKS: 80
TIME: 2 HOURS

Answers to this Paper must be written on the paper provided separately. You will not be allowed to write during first 15 minutes.

This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.

Section A is compulsory. Attempt any four questions from Section B.

The intended marks for questions or parts of questions are given in brackets [].

(This paper consists of seven printed pages)

SECTION A

(Attempt all questions from this Section)

Question 1

Choose the correct answers to the questions from the given options. [15]
(Do not copy the questions, write the correct answers only.)

- (i) Chile salt peter is heated with concentrated sulphuric acid at a temperature below 200°C . Which of the following could be the products:

P : Sodium sulphate.

Q : Nitric acid.

R : Hydrogen chloride gas.

S : Sodium hydrogen sulphate.

(a) both P and R (b) both Q and S (c) only P (d) only R

- (ii) A compound M in a test tube is reacted with dilute sulphuric acid. The gas evolved is bubbled through an aqueous solution of lead acetate and a black precipitate is obtained. Which of the following could be M?

(a) Ferrous chloride (b) Sodium sulphide (c) Zinc chloride (d) Magnesium chloride

- (iii) The relative atomic mass of nitrogen is 14, and that of hydrogen is 1. This means that

(i) _____ of nitrogen has the same mass as (ii) _____ of hydrogen.

	(i)	(ii)
(a)	An atom	28 molecules
(b)	An atom	7 molecules
(c)	A molecule	14 atoms
(d)	A molecule	7 atoms

- (iv) When conc. hydrochloric acid is reacted with a black solid, a greenish yellow gas with a pungent odour is released. Which property of the black solid is exhibited in this reaction?

(a) Reducing agent (b) Basic gas (c) Supporter of combustion (d) Oxidising agent

- (v) Assertion (A): A solution of ammonia gas in toluene has no effect on red litmus.

Reason (R) : Ammonia gas does not undergo ionization in toluene.

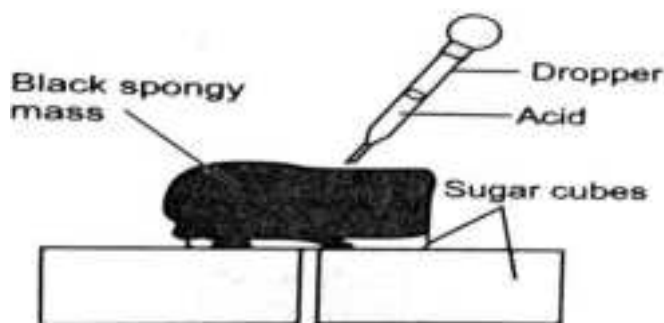
(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) Both A and R are false.

- (vi) Which of the following molecules has two lone pair of electrons and two bond pairs?
 (a) Carbon dioxide (b) Water (c) Ammonia (d) Nitrogen
- (vii) Element 'Z' has electronic configuration 2, 8, 3. The number of fluorine atoms present in the fluoride of 'Z' is:
 (a) 1 (b) 2 (c) 3 (d) 4
- (viii) $^{18}_8\text{O}$ is an isotope of oxygen. In the modern Periodic Table it will be placed:
 (a) before oxygen
 (b) after oxygen
 (c) at the same position as oxygen
 (d) not have any position in the Periodic Table
- (ix) A sulphate which forms a precipitate with sodium hydroxide and is soluble in excess of it:
 (a) Copper sulphate (b) Ferric sulphate (c) Zinc sulphate (d) Magnesium sulphate
- (x) Which of the following electronic configuration represents the MOST electropositive element?
 (a) 2, 1 (b) 2, 2 (c) 2, 8 (d) 2, 7
- (xi) Assertion (A): Noble gases do not form ions.
 Reason (R) : They have stable octet electronic configuration
 (a) Both A and R are true, and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true but R is false.
 (d) Both A and R are false.
- (xii) 10 g of magnesium carbonate reacts completely with excess dilute acid according to the equation: $\text{MgCO}_3 + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2\text{O} + \text{CO}_2$. What volume of carbon dioxide is formed? [O = 16, Mg = 24, C = 12]
 (a) 2.8 dm^3 (b) 2.6 dm^3 (c) 2.2 dm^3 (d) 2.4 dm^3
- (xiii) Concentrated and dilute nitric acid can be distinguished using:
 (a) Sodium (b) Copper (c) Magnesium (d) Iron
- (xiv) Which of the following about amphoteric oxides is INCORRECT?
 (a) Amphoteric oxides are insoluble water.
 (b) Amphoteric oxides are mostly ionic.
 (c) Amphoteric oxides are basic.
 (d) Amphoteric oxides are mostly oxides of non-metals.
- (xv) Arti added concentrated sulphuric acid to sugar cubes as shown in the figure below.
 Which property of sulphuric acid is demonstrated in the experiment set up shown below:

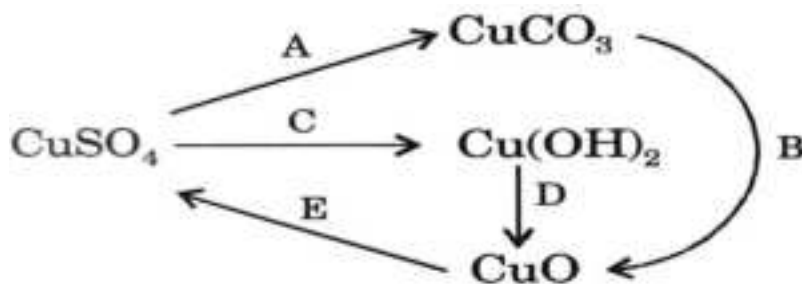


- (a) Non volatile acid (b) Oxidising agent
 (c) Dehydrating agent (d) Typical acid

Question 2

(i) Write balanced chemical equations for the conversions A to E:

[5]



(ii) Match the following Column A with Column B:

[5]

Column A	Column B
(a) Neon	1. Ammonium ion
(b) Methane	2. Hydronium ion
(c) Ammonia gas in water	3. Ammonium hydroxide
(d) An ion with a lone pair of electrons	4. Non polar covalent compound.
(e) A weak alkali	5. Zero electron affinity

(iii) Complete the following sentences by choosing the correct answers from the bracket:

[5]

- If an element has seven electrons in the valence shell, then it is likely to have the _____ [smallest/ largest] atomic radius amongst all the elements in that same period.
- _____ [sulphuric acid/ hydrochloric acid] does not form an acid salt.
- A _____ [pale blue/ dirty green] coloured precipitate is formed when sodium hydroxide is added to a solution of ferrous chloride.
- Calcium chloride undergoes _____ [ionization/ dissociation] in molten state.
- An _____ [acidic/ alkaline] solution will turn methyl orange solution pink.

(iv) Write the suitable term for the following statements:

[5]

- A pair of electrons which does not take part in covalent bond formation.
- A salt formed by incomplete neutralization of a base by an acid.
- The negative logarithm of H^+ ion concentration in moles per litre to the base of ten.
- The amount of energy released when a neutral isolated gaseous atom accepts an electron in its outermost shell.
- The volume occupied by one mole of a gas at STP.

(v) (a) Draw the electron dot and cross diagram for the following:

[5]

- Methane
- Hydronium ion
- Magnesium oxide

(b) Define the following:

- Mole
- Ionisation potential

SECTION B (40 Marks)

(Attempt any four questions)

Question 3

(i) Identify the reactant 'M' and write a balanced chemical equation for the following:

[2]

Hydrochloric acid reacts with compound 'M' to give a blue solution, water and carbon dioxide

(ii) What property of sulphuric acid is exhibited in each of the following cases: [2]

- (a) In the preparation of nitric acid when it reacts with sodium nitrate.
(b) Dilute sulphuric acid reacts with Magnesium to produce Hydrogen.

(iii) Study the portion of periodic table given below:

Alphabets A to L do not represent the symbols of elements.

[3]

Group →	IA	IIA	IIIA	IVA	VA	VIA	VIIA	VIIIA
	Li		D			O	J	Ne
	A	Mg	E	Si		H	K	
	B	C		F	G			L

- (a) How is the reducing power of A likely to compare with that of H?
(b) How is the atomic radius of F likely to compare with that of G?
(c) Compare the electron affinity of Li and B.

(iv) You are provided with the list of chemicals mentioned below in the box: [3]

Nitric acid, potassium hydroxide solution, copper carbonate, zinc, copper, dilute hydrochloric acid, silver nitrate, sulphuric acid, chlorine, copper hydroxide.

Using suitable chemicals from the list given, write balanced chemical equation for the preparation of salts mentioned below: [3]

- (a) Copper sulphate
(b) Potassium zincate
(c) Silver chloride

Question 4

(i) The following questions relate to the industrial preparation of ammonia: [2]

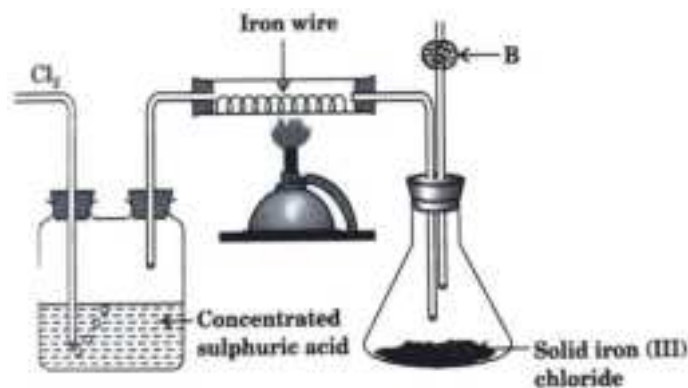
- (a) Write a balanced chemical equation for the catalyzed reaction.
(b) How is ammonia gas separated from the unused reactants?

(ii) On analyzing an organic sample weighing 10.47 grams, Rahul found the sample contains 6.21 grams of lead (atomic weight 207) and 4.26 gram of chlorine (atomic weight 35.5). Determine the empirical formula of the compound. [2]

(iii) Give balanced chemical equation for each of the following: [3]

- (a) Action of warm water on calcium nitride.
(b) Oxidation of sulphur with nitric acid.
(c) Excess ammonia is reacted with chlorine.

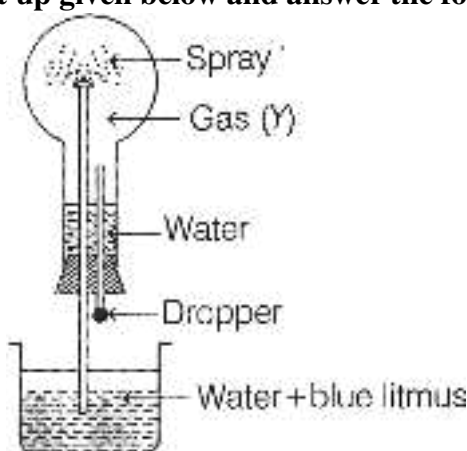
(iv) The diagram given below is to prepare ferric chloride in the laboratory: [3]



- (a) Name the method for preparation of ferric chloride.
(b) Write balanced chemical equation for the preparation of ferric chloride.
(c) What would you observe if caustic soda solution is added to a solution of ferric chloride?

Question 5

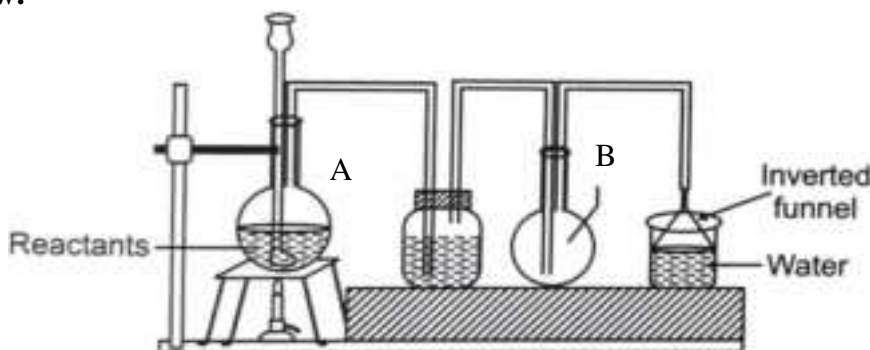
- (i) (a) Aparna passed a mixture of dry ammonia gas and oxygen gas over heated platinum at a temperature of 700°C . What did she observe?
(b) Write a balanced chemical equation for the above reaction. [2]
- (ii) Arrange the following as instructed: [2]
(a) Methane, water, ammonia (decreasing order of number of single bonds)
(b) Nitrogen, boron, fluorine (increasing order of number of valence electrons)
- (iii) Rohit was given a salt 'X' for analysis which was white in colour. On strong heating it produced a residue which is yellow when hot and white on cold, a colourless gas and a reddish brown gas.
The solution of salt 'X' when tested with excess of ammonium hydroxide produced a colourless solution.
(a) Name the coloured gas evolved when Rohit heated the salt strongly.
(b) Which cation was present in the salt given to Rohit?
(c) Identify the salt given to Rohit for analysis. [3]
- (iv) Study the experimental set up given below and answer the following questions: [3]



- (a) Identify gas Y.
(b) Name the experiment.
(c) Which MAIN property of gas Y is demonstrated in this experiment?

Question 6

- (i) Element 'Y' forms an oxide with formula YO , which is a solid with high melting point. 'Y' would most likely be placed in the group of the Periodic Table as:
1. (a) Li (b) Ca (c) B (d) C
2. Justify your answer in the above question (1). [2]
- (ii) A student prepared hydrochloric acid in the laboratory by setting up the experiment given below: [2]



- (a) Write balanced chemical equation for the reaction taking place in the round bottom Flask 'A'.
(b) What is the role of the inverted funnel?

- (iii) The gases hydrogen, nitrogen dioxide, carbon dioxide and oxygen are collected under the same conditions of temperature and pressure. The following table gives the volumes of gases collected and the number of molecules (X) in 20 litres of nitrogen. Copy and complete the following table giving the number of molecules in the other gases in terms of X [3]

Gas	Volume (litres)	Number of molecules
Hydrogen	10	
Nitrogen dioxide	20	X
Carbon dioxide	20	
Oxygen	5	

- (iv) X [2, 8, 2] and Y [2, 8, 1]. Using this information complete the following: [3]
- _____ is the alkaline earth metal.
 - The element with lower ionization potential is _____.
 - _____ is the oxidizing agent.

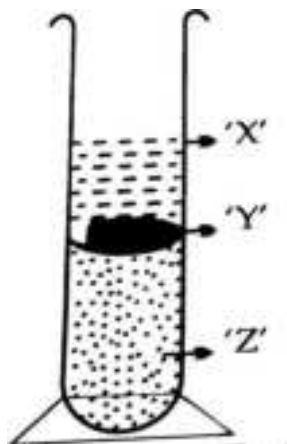
Question 7

- (i) 200 cc of ethyne is burnt in just sufficient air (containing 20% oxygen) to form carbon dioxide gas and steam.
The equation for the reaction is: $2\text{C}_2\text{H}_2(\text{g}) + 5\text{O}_2(\text{g}) \rightarrow 4\text{CO}_2(\text{g}) + 2\text{H}_2\text{O}$
Calculate : (a) The volume of oxygen used.
(b) The volume of air needed.
(c) The volume of carbon dioxide formed. [3]
- (ii) Raman has three solutions X, Y and Z having pH values as 12, 7 and 1 respectively. Which solution: [3]
- will liberate sulphur dioxide, when added to sodium sulphite.
 - will turn pH paper green.
 - will produce a pungent smelling basic gas when heated with ammonium sulphate.
- (iii) Raghav heated 8.2 g of calcium nitrate in a clean and dry test tube. The equation for the reaction is: $2\text{Ca}(\text{NO}_3)_2 \rightarrow 2\text{CaO} + 4\text{NO}_2 + \text{O}_2$
Calculate the following:
- Volume of nitrogen dioxide produced.
 - Mass of calcium oxide formed.
 - Volume of oxygen formed.
 - Number of molecules present in 8.2 g of calcium nitrate. [Ca = 40, O = 16, N = 14] [4]

Question 8

- (i) Give reason for the following: [2]
- Magnesium nitride is not used for the laboratory preparation of ammonia.
 - Sodium hydroxide solution cannot be used to distinguish between lead nitrate and zinc nitrate solution.
- (ii) Compare the properties of magnesium chloride and carbon tetrachloride based on: [2]
- Solubility in water.
 - Electrical conductivity
- (iii) Identify the salts P, Q, R from the following observations: [3]
- Salt P has is a green solid. On heating leaves a black residue. Salt P produces a colourless, odourless gas with brisk effervescence on addition of dilute sulphuric acid, the produced gas turns lime water milky and has no effect on acidified potassium dichromate solution.
 - Salt Q is a white solid which gives a non-persistent brick red coloured flame. A curdy white precipitate is formed on adding silver nitrate solution to an aqueous solution of Q.

- (c) Salt R is a white crystalline solid which imparts a golden yellow colour to the flame. On adding barium chloride solution to an aqueous solution of R a dense white precipitate is formed insoluble in mineral acids.
- (iv) Class X students performed the Brown Ring Test using a metallic nitrate as shown in the figure below. Answer the questions given [3]



- (a) Identify Z.
- (b) Write the formula of Y.
- (c) Why is freshly prepared ferrous sulphate solution used in this test.